
Management Team

INFILTR8
Software Requirements Specification
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The Guidance Team and the customer will approve this document.

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Change Summary

The following table details changes made between versions of this document

Version	Date	Modifier	Description
0.1		Guidance team	Create main section and section 1 content
0.2		Guidance team	Create section 2 and UC diagram
0.3		Guidance team	Real World Requirements and Class diagram
0.4		Guidance team	User interfaces
0.5		Guidance team	Behavioral Requirements and STD
0.6	7/18/2024	Guidance team	Data Flow Diagram and Data Dictionary
0.7	7/20/2024	Guidance team	Consistency checking among models
0.9	8/19/2024	Sofia Holguin	Modifying the requirements of the ML component

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1.0	08/26/24	Sebastian Quinones	Major Modifications to the Core Functionality. Included dependency for DAC generated components. Modified Frontend and Backend developmental requirements.
1.1	08/29/24	Sebastian Quinones	Final Revision to transform pentest into analysis

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1. Introduction

1.1. Purpose and Intended Audience

The *Software Requirements Specification* (SRS) aims to provide the analyst with a detailed description of how to use the software created for the *Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance* (INFILTR8) system. The *Data Analysis Center's* (DAC) Analyst (the analyst from now on) is the main user of the INFILTR8 system. The software system allows the analyst to ingest reconnaissance files and analyze them using an *Artificial Intelligence* (AI) component of the INFILTR8 system. That AI component is a *Machine Learning* (ML) algorithm that analyzes the reconnaissance files and generates a prioritized list of entrypoints for the analyst to refer to during the penetration testing. The analyst can specify the entry point types to analyze on the allowed devices to determine their safety. The entry point types include unauthorized port bypass, default credentials, unpatched software exploits, protocols missing encryption, and weak passwords; each of these entrypoints is intended to be indicative of a different type of vulnerability present within the scan. INFILTR8 facilitates the execution of the penetration testing process to DAC Analysts. This SRS is intended for analysts who seek to benefit from using our tool in the field when conducting penetration tests, the designers working on this tool's creation, and all other project stakeholders.

1.2. Scope of the Product

The *Department of Defense* (DoD) must ensure that the technological systems they are delivering have secure and resilient capabilities. To ensure these capabilities, and the survivability and lethality of the combat systems used, it is critical for them to perform security testing of these systems. During this phase of testing, the analyst, acting as a penetration tester, is responsible for the discovery of vulnerabilities and realizing the exploitation tied to the vulnerabilities. Then, analysts generate a report describing the risks of each vulnerability. The product being developed is named *Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance* or INFILTR8. Currently, analysts are required to perform all steps of a penetration test manually, with little to no automation. The INFILTR8 system shall give the analyst a way to facilitate and expedite the penetration testing process. This shall be done by ingesting reconnaissance files and developing a prioritized list of entrypoints within the set of machines that are on the network.

The *Cyber Experimentation & Analysis Division* (CEAD) needs a tool to allow analysts to expedite the beginning stages of penetration testing and aid the analysts in visualizing the progress of the undergoing penetration tests. The *System of Interest* (SOI), INFILTR8, provides the following functionality:

- a. Ingest reconnaissance files from external tools.
- b. Compile the reconnaissance results into a format the SOI can use.
- c. Use the ML component of the SOI to generate a ranked entrypoint list from the reconnaissance files.
- d. Generate a new project based on the scope set by the analysts, including scheduling an analysis and selecting the list of allowed devices on which the analysis shall be performed.
- e. Allow the analyst to select the type of entrypoints that will be analyzed from the reconnaissance results.
- f. Display the current progress of the analysis to the analyst.
- g. Allow the analyst to go back to setting the scope of the analysis when new devices need to be added to the current test.
- h. Generate a file to export the compiled results of the analysis.

INFILTR8 does **not** do the following:

- a. Perform the reconnaissance needed for the project.
- b. Modify the ingested reconnaissance files during the ingestion process.
- c. Actively perform adversarial actions on the devices within the network

INFILTR8 benefits include:

- a. Provide the CEAD analysts with the required tools to analyze and prioritize the vulnerabilities of the system.
- b. Provide a detailed report of the ranked entry points found.
- c. Create a more streamlined analysis of the ingested logs.

INFILTR8 goals include:

- a. Improve observational reporting.
- b. Improve workflow efficiency.

1.3. Definitions, Acronyms, and Abbreviations

1.3.1. Definitions

Table 1.3.1-1: Definitions

TERM	DEFINITION
Valid Entrypoints	Entrypoints that are available on the network found in the result files from the Nessus reconnaissance tool.
CAT 1 Vulnerabilities	Vulnerabilities that will directly and immediately result in loss of confidentiality, availability, or integrity. They can allow unauthorized access to classified data or facilities. These vulnerabilities are the most severe.
Cyber Vulnerabilities	The weaknesses in a system that an attacker can use to gain access to the system.
Project	An instance of the INFILTR8 system that contains the information of one assessment of the allowed devices.
Analyst (DAC Analyst)	The Data Analysis Center's (DAC) Analyst is the main user of the INFILTR8 system. DAC Analyst perform penetration testing. INFILTR8 facilitates the execution of penetration testing to DAC Analysts.
Textual File	A file that is made up of pure text only.

1.3.2. Acronyms

This section lists the acronyms used in this document and their associated definitions.

Table 1.3.2-1: Acronyms

TERM	DEFINITION
INFILTR8	Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance (INFILTR8) system
SRS	Software Requirements Specification
SOI	System of Interest
DAC	Data Analysis Center
STD	State Transition Diagram
CD	Class Diagram
DFD	Data Flow Diagram
DD	Data Dictionary
CAT	Category
CEAD	Cyber Experimentation & Analysis Division
DEVCOM	U.S. Army Combat Capabilities Development Command

DoD	Department of Defense
IP	Internet Protocol
Recon	Reconnaissance
ML	Machine Learning

1.3.3. Abbreviations

Table 1.3.3-1: Abbreviations

TERM	DEFINITION
e.g.	For example
UI	User interface - The part of the system with which the user interacts with directly
DB	Database - An organized collection of data

1.4. Overview

The Software Requirements Specification (SRS) document provides the analysts and any other stakeholders with a detailed description of the INFILTR8 system, particularly the system's intended functionality, and the mechanisms of user interaction. The SRS document has a predefined outline. A brief synopsis of what each section covers follows (please note that the numbering corresponds to the section number in the SRS):

1. **Introduction:** Briefly introduce the system to be developed, including its purpose and scope. The main terms, acronyms, and abbreviations of the SRS are further defined.
 - 1.1. **Purpose and Intended Audience:** Highlights the purpose and audience of the SRS.
 - 1.2. **Scope of Product:** Describes the purpose, goals, benefits, and background of the system.
 - 1.3. **Definitions, Acronyms, and Abbreviations:** Defines terms, acronyms, and abbreviations used in the report.
2. **General Description:** Provides an overall view of the system.
 - 2.1. The **Product Perspective** section describes how the system fits into the company's overall structure and its relationship with other systems.
 - 2.2. The **Product Features** provides a brief description of all the system's main features. This section contains the Use Case Diagram, a description of Actors, and Use Cases that define the user's interactions with the system.
 - 2.3. The **User Characteristics** section describes the intended users of the system and any characteristics that may influence the system design.
 - 2.4. The **General Constraints** section identifies any limitations that may impact the development of the software.
 - 2.5. The **Assumptions and Dependencies** section states any assumptions made during the development of the requirements and identifies external factors that may affect the system.
3. **Specific Requirements:** Delves into the detailed technical requirements, such as external interfaces, behavioral, and non-behavioral that allow the design of the system to satisfy all the conditions described in the Requirements Definition Document (RDD).
 - 3.1. **External Interface Requirements:** Describes the interface between the system and external entities, such as users, hardware, or other systems.
 - 3.2. **Behavioral Requirements:** Specifies the expected behavior of the system under various circumstances.
 - 3.3. **Non-Behavioral Requirements:** Covers requirements for the quality attributes of the system, such as performance, usability, and security requirements.
 - 3.4. **Other Requirements:** Describes requirements mostly pertaining to data that flows through the system and how it is transformed.

1.5. References

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2. General Description

2.1. Product Perspective

[Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance \(INFILTR8\)](#) is a cybersecurity tool designed to assess the existing vulnerabilities on a set of computer systems and help facilitate and expedite the penetration testing process. INFILTR8 is an independent, self-contained software system designed to assist analysts in performing decisions during a penetration test. It shall offer a user-friendly interface to help the analyst categorize and prioritize potential vulnerabilities within the designated IP addresses. The INFILTR8 system shall leverage a Machine Learning algorithm to process reconnaissance files and generate a ranked list of exploitable entry points. The analyst can then use this list to determine what systems should be tested first during a penetration test. INFILTR8 assists the analysts in expediting the penetration testing process and will be an essential tool to add to the company's toolset.

[Currently, Cyber Experimentation & Analysis Division \(CEAD\)](#) analysts perform all steps of a penetration test manually. INFILTR8 addresses CEAD's need to generate a list of endpoints to expedite the initial stages of the penetration test. INFILTR8 is designed as an independent solution within CEAD analysts' work environment and will ingest reconnaissance files and leverage an AI component to optimize an endpoint list to supplement further penetration testing efforts for the analysts. INFILTR8 creates an efficient environment that enables the analyst to generate the endpoint list and use that endpoint list to perform the penetration test on the allowed devices.

2.2. Product Features

The main features of this SOI include the following functionalities: Managing a project, setting the scope of the project, managing the penetration testing, and logging the activity. These features are shown in the Use Case Diagram - Level 1, see Figure 2.2.3-1. The actors (e.g., users, devices, other systems) for the SOI are the DAC analysts and the allowed devices on which the penetration will be performed. The functionalities shown in Figure 2.2.3-1 will be expanded upon by describing them in three Level 2 Use Case diagrams.

2.2.1. List of Actors with Descriptions

The main actors of the INFILTR8 system are the following:

Actor	Description
Analyst	A cyber analyst working for the Data Analysis Center's Cyber Experimentation & Analysis Division (DAC CEAD) will operate the Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance (INFILTR8) system. The analyst can use the INFILTR8 to create a project, process reconnaissance results, and execute an analysis. Each analyst has full access to the system once they have provided a valid login.

2.2.2. List of Use Cases with Descriptions

2.2.2.1 Level 1 Use Cases

Use cases are grouped by levels to cope with the complexity of the INFILTR8 system. The Use Case Level 1 includes abstract use cases identified by separations of concerns. The following four Use Cases belong to Level 1 and are further decomposed into a Level 2 Diagram.

Use Case	Description
Manage Project	The analyst can manage projects by creating a new project, processing reconnaissance results, and reviewing the prioritized list of entrypoints provided by the Intelligent Network Findings with In-Depth Learning for Targeted Reconnaissance (INFILTR8) system. As seen in Figure 2.2.3-1 and elaborated in Figure 2.2.3-2.
Set Scope	The analyst can set the project's scope by specifying the list of IPs (allowed and disallowed devices) and types of attacks and scheduling an attack. As seen in Figure 2.2.3-1 and elaborated in Figure 2.2.3-3.
Manage Penetration Testing	The analyst can manage the analysis by executing the AI algorithm, displaying progress, exporting results, and viewing results. As seen in Figure 2.2.3-1 and elaborated in Figure 2.2.3-4.
Log Activity	The analyst can view a log for either user actions or system actions. As seen in Figure 2.2.3-1.

2.2.2.2 Level 2 Use Case

Level 2 Use Cases for INFILTR8 are organized by separation of concerns identified in level 1: Manage Project, Set Scope, Manage Penetration Testing, and Log Activity:

Manage Project:

Use Case	Description
Login	The analyst logs in to the INFILTR8 system before opening or creating a project.
Open Project	The analyst opens an existing project to resume the analysis.
Create Project	The analyst creates a new project to manage an analysis event and provides the name and date of the event.
Process Reconnaissance Results	The analyst can select a path where the reconnaissance result's files are located. Then INFILTR8 ingests the selected files, analyzes the data, and creates a ranked list of valid entry points and their associated exploit(s) using the Machine Learning algorithm. Processing reconnaissance files allows the analyst to recognize the context of the environment and the extent of the Artificial Intelligence capabilities. The reconnaissance files ingested shall be provided in the form of Nessus Files.

Set Scope:

Use Case	Description
Schedule Analysis	The analyst can schedule an Analysis at a particular time of the analyst's choosing. This includes beginning the test at a designated time or ending by a specified time.
Specify List of IPs	The analyst can specify the list of the devices' IPs to establish which IPs are allowed and which are off-limits.
Set Scope: Type of attacks	The analyst can set the scope of the entry point types the system can analyze during the ingestion. This includes allowing certain attacks or disallowing attacks from the analysis. The entry point types available to the analyst are unauthorized port bypass, default credentials (specific services), unpatched software exploits, protocols missing encryption (detecting and using password exchanges), and weak passwords (brute force).

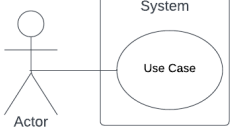
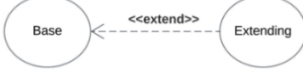
Manage Penetration Testing:

Use Case	Description
Execute analysis	The analyst can start the execution of the analysis based on the analyst's selection. If the analyst does not select a type of entry point, the system shall analyze all available entry point types. The valid entry points available are unauthorized port bypass, default credentials (specific services), unpatched software exploits, protocols missing encryption (detecting and using password exchanges), and weak passwords (brute force). The analyst can pause, resume, and cancel a currently ongoing analysis. The analyst may go back to the project's creation if there are new allowed devices to include in the analysis.
Display Progress	The analyst can display/hide the progress of the analysis as it proceeds.
Export Analysis Results	The analyst can export the results as a ranked list of recommended entry points from the analysis as an XML file or a PDF file.
View Analysis Results	The analyst can view the results from the completed analysis. The results will be shown to the analyst in a weighted list of entry points to provide the analyst with a clear way to proceed.

2.2.3. Use Case Diagrams

The following section details the use case diagrams for INFILTR8. Table 2.2.3-1 provides a legend for the use case diagrams in this section.

Table 2.2.3-1: Use Case Diagram Legend

NAME	NOTATION	DESCRIPTION
System		The rectangle represents the boundary of the system. It separates the system from its external environment.
Use Case		Service provided by the system.
Actor		An external entity that interacts with the system via the use cases. Role of the users of the system.
Association		The actor participates in the Use Case.
Include Relationship		The Base use case includes the Included use case. The Base use case requires the behavior of the Included use case to provide its functionality.
Extend Relationship		The Base use case extends the Extending use case. The Base use case can optionally use the behavior of the Extending use case.

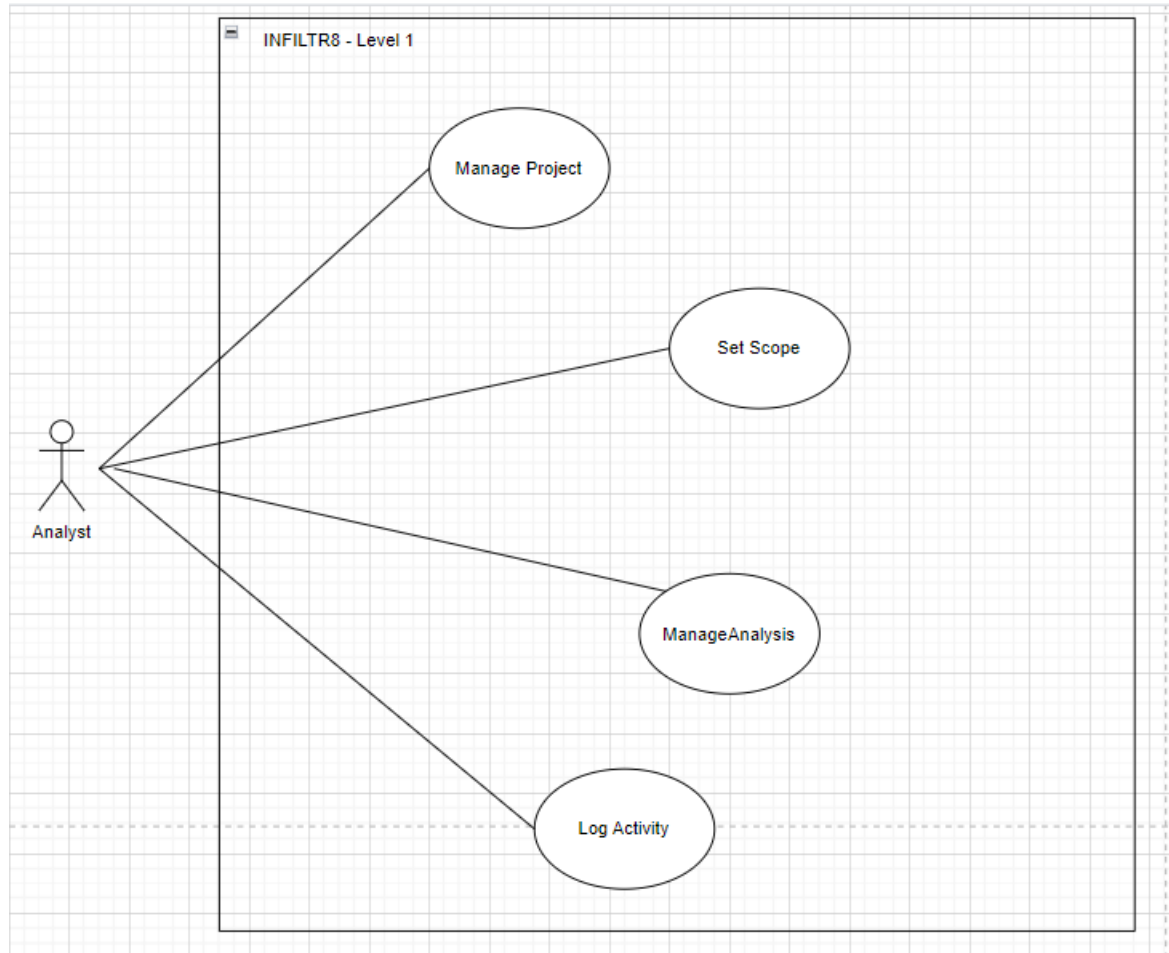


Figure 2.2.3-1. INFILTR8 Use Case Level 1

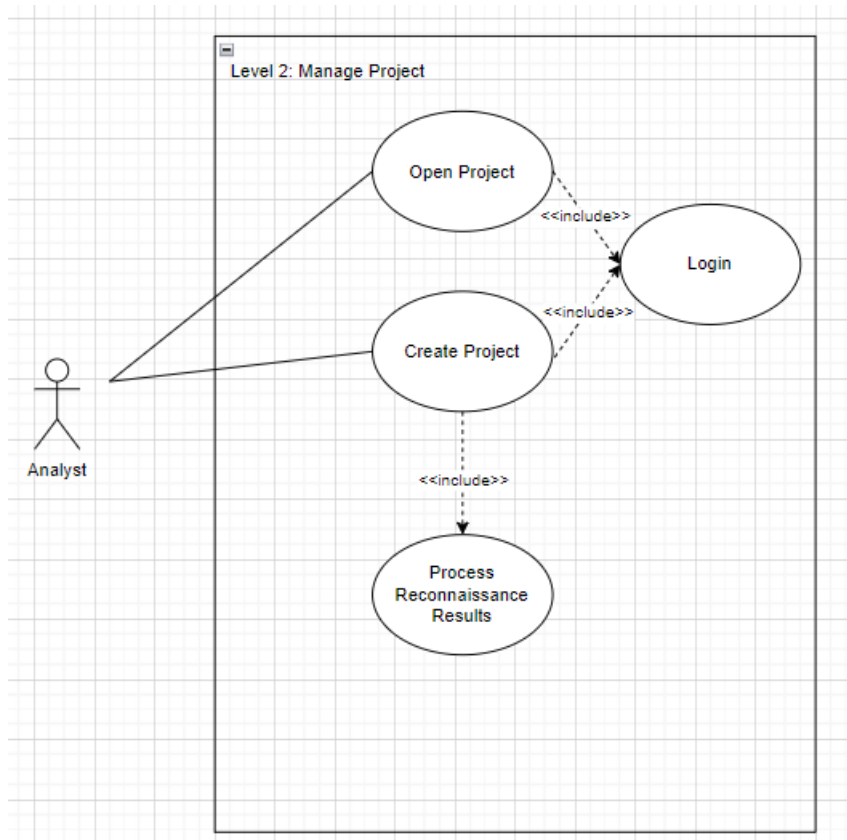


Figure 2.2.3-2 Level 2 Manage Project

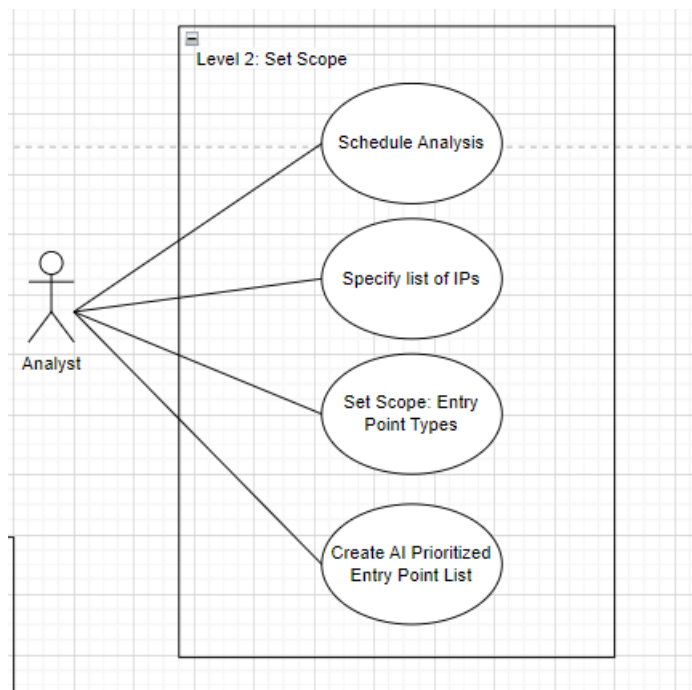


Figure 2.2.3-3. Level 2 Set Scope

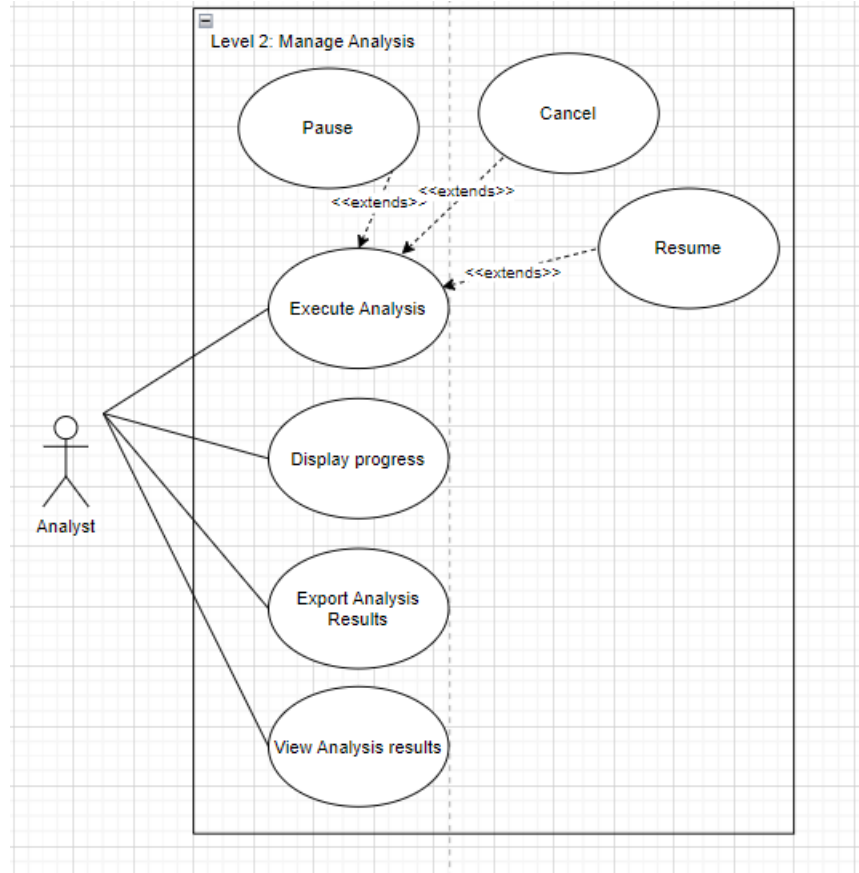


Figure 2.2.3-4. Level 2 Manage Analysis

2.3. User Characteristics

This section identifies the users of the system, the nature of their use of the system, and their characteristics.

UC1. Users shall be authorized members of the DEVCOM Analysis Center who are granted access to the system via login credentials.

UC2. The users may have disabilities that prevent them from using the system

UC3. The system shall provide accommodations such as color blindness to aid the analyst in using the tool better.

UC4. The users shall be expected to have some experience with the penetration testing process.

2.4. General Constraints

GC1. The system shall be a web-based application.

GC2. The system shall be capable of running smoothly on a machine with the specifications provided on the RDD: 4GB Ram, 4-Core 2.5Ghz, and 20GB of available storage.

GC3. The computer where INFILTR8 is installed will have a Chrome or Firefox browser installed.

GC4. The computer where INFILTR8 is installed will have a Kali Linux Operating System.

2.5. Assumptions and Dependencies

Assumptions:

A1. The analyst, the user of the SOI, is expected to have experience using similar tools and how to interpret the results provided by the system for their gain.

A2. The SOI will allowed users within the intranet to login to the application if they provide the correct credentials.

Dependencies:

D1. CEAD will provide foundational functionality for the analysis aspect of INFILTR8. The code provided should be considered in a “beta” stage and NOT FEATURE COMPLETE; to be delivered on or before September 3rd, 2024.

D2. CEAD will provide at least one (1) Nessus scan so that development teams can dynamically test their application on or before September 3rd, 2024.

2. Specific Requirements

3.1. External Interface Requirements

This section contains the specification of requirements for interfaces among different components and their external capabilities, including all its users, both human and other systems. The characteristics of interfaces to SOI are included. Each interface may represent a bi-directional flow of orientation.

3.1.1. User Interfaces

This section addresses the main user interfaces for INFILTR8 and their corresponding GUI components.

NOTE: The following user interfaces are only potential options that may be considered for development and these user interfaces are not required.

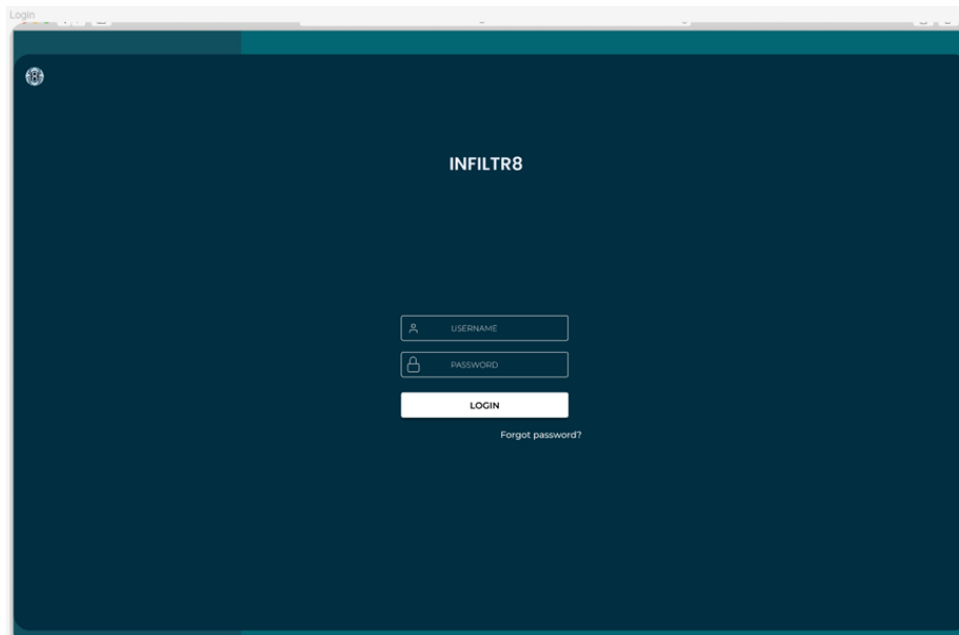


Figure 3.1.1-1 Login screen

[SRS 3.1.1.1] INFILTR8 shall contain a “Login” screen that allows a user to get access to the platform. The login screen is shown in Figure 5.

[SRS 3.1.1.2] The login screen shall include the components:

- a. The name of the system.
- b. A button labeled “Login”.
- c. A button labeled “Forgot Password?”.
- d. A textbox labeled “Username”.
- e. A textbox labeled “Password”.

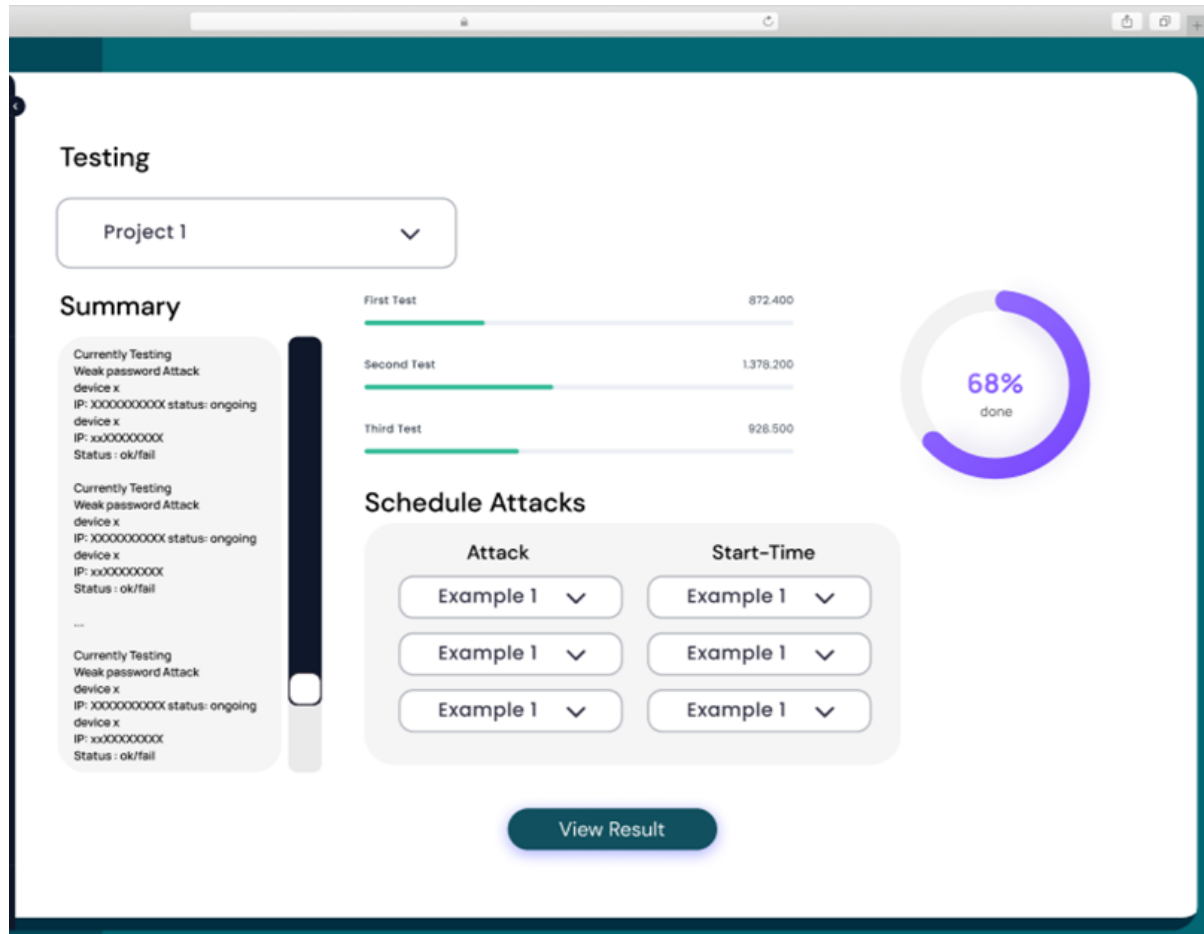


Figure 3.1.1-2. Testing Screen

[SRS 3.1.1.3] INFILTR8 shall contain an “Analysis” screen that displays in real time the list of currently running analysis with details on its process, allows scheduling an analysis, and allows selecting a project to view. The Analysis screen is shown in Figure 6.

[SRS 3.1.1.4] The analysis screen shall include the components:

- a. A page title label “Analysis”.
- b. A drop box to select the project currently being run.
- c. A summary section to display the Analysis being run.
- d. A scroll bar to navigate through the summary section.
- e. Progress bars for the Analysis being run.
- f. A label for each of the progress bars stating which analysis is being run.
- g. A progress bar to display the overall progress of the project.
- h. A label stating the percentage of the analysis completed.
- i. A label for the section “Schedule Analysis”.
- j. A label “Entry Points” for the selection of the type of entry points.
- k. A drop box providing the options of the entry point types.
- l. A label “Start-Time” for the selection of the time when the analysis will take place.
- m. A drop box providing the available options for the start time.
- n. A button labeled “View Result”.

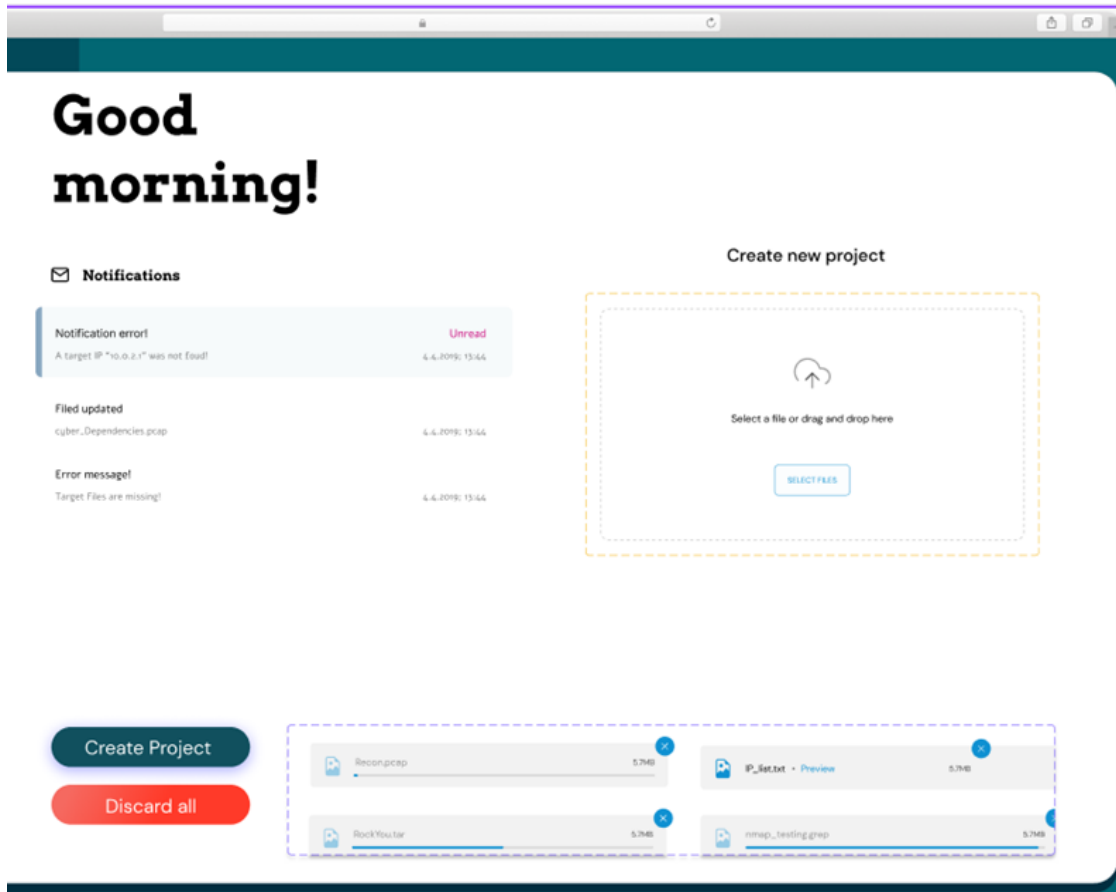


Figure 3.1.1-3. Dashboard Screen.

[SRS 3.1.1.5] INFLTR8 shall contain a “Dashboard” landing page, that provides project alerts, and allows users to create a new project. The Dashboard screen is shown in Figure 7.

[SRS 3.1.1.6] The Dashboard screen shall include the components:

- a. A label to greet the analyst depending on the time of day.
- b. A label “Notifications” for the section of notifications to the analyst.
- c. A list of notifications.
- d. A label “Create New Project”.
- e. A section to select file or drag and drop a file.
- f. A button labeled “Select Files”.
- g. A button labeled “Create Project”.
- h. A button labeled “Discard all”.
- i. A section displaying the files being uploaded and their progress.

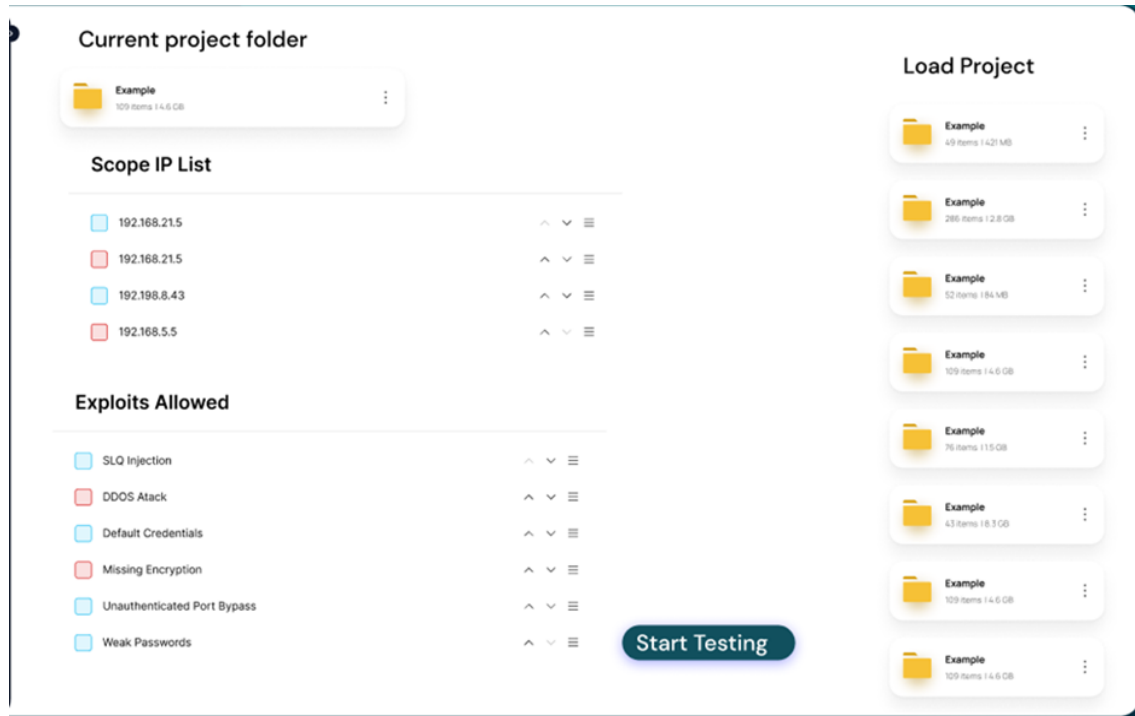


Figure 3.1.1-4. Project Screen.

[SRS 3.1.1.7] INFILTR8 shall also have a “Project” screen that allows the analyst to set the scope IP list and select the folders of the project they need to run. The Projects screen is shown in Figure 8.

[SRS 3.1.1.8] The Dashboard screen shall include the components:

- a. A page title label “Current project folder”.
- b. A button to the current project folder.
- c. A label “Scope IP List”
- d. A list of IPs.
- e. An up arrow next to each IP to be able to move it up on the list.
- f. A down arrow next to each IP to be able to move it down on the list.
- g. A label “Entry Points Allowed”.
- h. A list of all available analyses.
- i. An up arrow next to each test name to be able to move it up on the list.
- j. A down arrow next to each test name to be able to move it down on the list.
- k. A button labeled “Start Analysis”.
- l. A label “Load Projects”.
- m. A list of projects available to load.

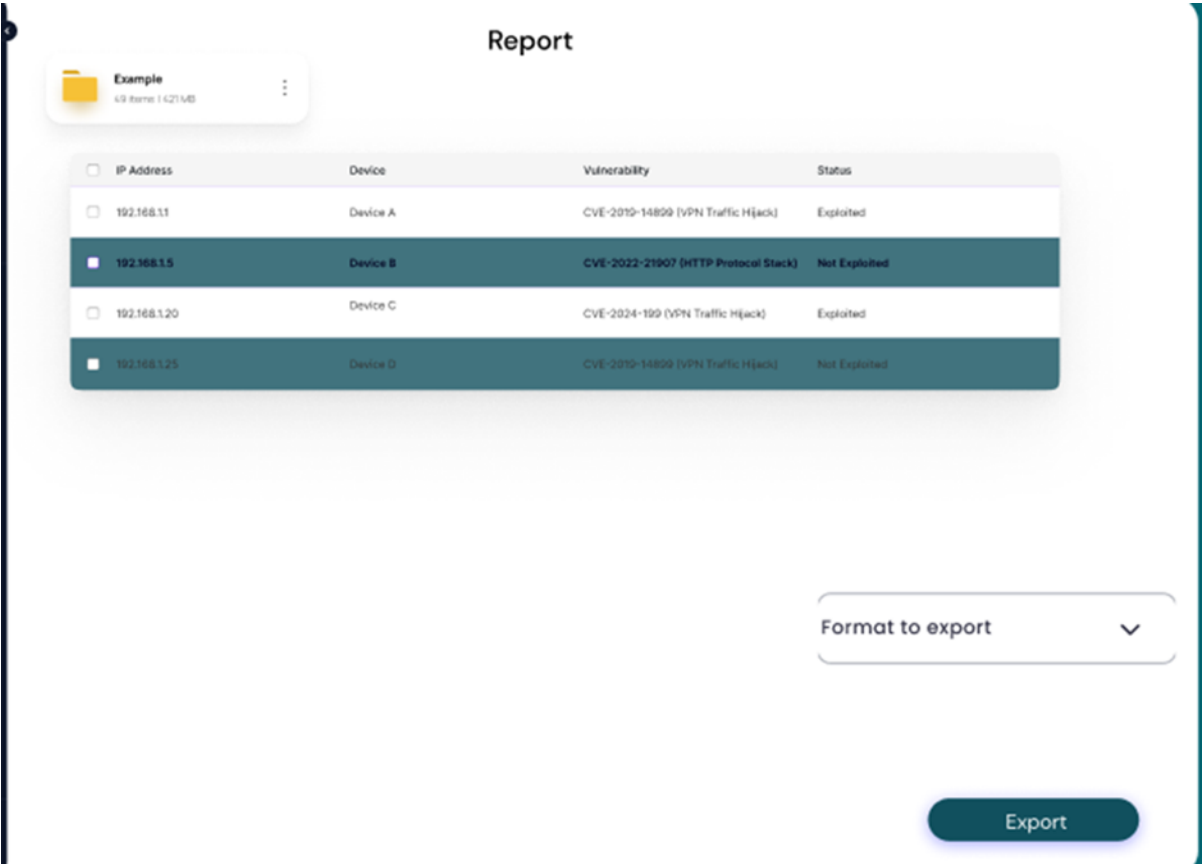


Figure 3.1.1-5. Report Screen

[SRS 3.1.1.9] INFILTR8 component interface shall have a dedicated screen called “Reports” displaying the current project with options to select which project to view and export results if that project is completed. The Report screen is shown in Figure 9.

[SRS 3.1.1.10] The Reports screen shall include the following components:

- a. A label “Report”.
- b. A button to the current project folder.
- c. A list containing the labels “IP Addresses”, “Device”, “Vulnerability”, and “Status”
- d. A list of devices with the corresponding information to the list mentioned in point C.
- e. A drop box labeled “Format to export”.
- f. A button labeled “Export”.

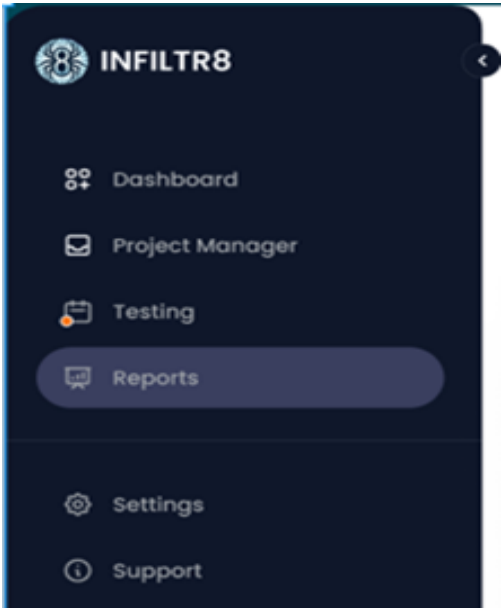


Figure 3.1.1-6. Collapsible Menu

[SRS 3.1.1.11] INFILTR8 interface shall incorporate a collapsible clickable menu of options on the left-hand side of the screen that allows the user to navigate to different screens and functionalities of the INFILTR8 component.

[SRS 3.1.1.12] This clickable menu, shown in Figure 10, shall be displayed on all screens to access it at any moment.

[SRS 3.1.1.13] The clickable menu shall include the components:

- a. A label “INFILTR8” for the name of the system.
- b. A button labeled “Dashboard”.
- c. A button labeled “Project Manager”.
- d. A button labeled “Analysis”.
- e. A button labeled “Reports”.
- f. A button labeled “Settings”.
- g. A button labeled “Support”.

3.1.2. Hardware Interfaces

Storage Devices: The system may require additional storage devices to store project files, such as project’s logs and project results.

3.1.3. Software Interfaces

INFILTR8 does not interact with any other software system. Analysts are expected to upload the results of Nessus scans into a folder on the same computer where the INFILTR8 system is running.

3.1.4. Communications Interfaces

[SRS 3.1.4.1] INFILTR8 shall transmit information by Secure Sockets Layer and Transport Layer Security (SSL/TLS) as stated in CAT 1 Vulnerability, V-222596.

3.2. Behavioral Requirements

3.2.1. Same Class of User

[SRS 3.2.1.1] INFILTR8 shall have a single level of access privilege assigned as "Analyst" with full access to system functionality.

3.2.2. Related Real-world Objects

Real-world objects represent business elements/components that interact to provide the services of a system of interest. Table 4 shows INFILTR8's real-world objects (business objects) and their purpose.

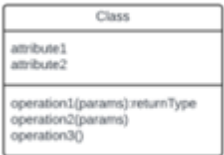
Table 3.2.2-1: INFILTRAT8 Real World Objects.

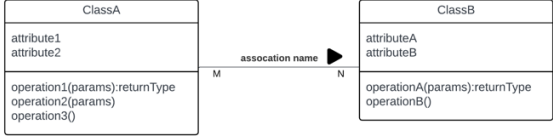
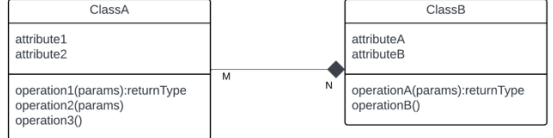
Real-World Object Name	Purpose
Scope Manager	The scope manager class allows the analyst to set the entry points allowed and set the scope of the allowed IPs.
Project Manager	The project manager is to create a project. The Project Manager class authenticates the analyst, retrieves the scope information, notifies the analyst that the scope is set, and requests the reconnaissance files.
Recon Result Manager	The Recon Result Manager ingests and verifies the validity of the reconnaissance files uploaded by the analyst and establishes a list of entrypoints.
Log Manager	The Log Manager creates one log for the actions and decisions of the system and another log for the analyst's actions and logs the actions taken throughout the lifetime of the project.
Analysis Scheduler	The purpose of the analysis scheduler is to give the analyst the option to schedule an analysis. The class provides notification when scheduling a analysis.
Result Manager	The Result Manager class allows the analyst to compile the analysis results, export the results to the analyst in their selected format, and store the project results.

The relations among real-world objects for this system are shown in Figure 12.

The purpose of the class diagram is to show the interaction among real-world objects to facilitate an understanding of how a system's services can be managed from a business point of view. The class diagram must be revisited when design constraints, quality attributes, and business constraints are considered during the architecture and detail design so relations and responsibilities may change, and new implementation classes can be added. The legend of the class diagram is shown in Table 3.2.2-2: Class Diagram Legend.

Table 3.2.2-2: Class Diagram Legend

NAME	NOTATION	DESCRIPTION
Class		A component of the system with its attributes (things it knows) and operations (things it does)
Unidirectional Association		ClassA is aware of ClassB and its relationship to ClassB. ClassB is not aware of ClassA.

Bidirectional Association		ClassA and ClassB are aware of each other and the relationship between them. ClassA is associated with N instances of ClassB. ClassB is associated with M instances of ClassA.
Composition		ClassA is strongly a part of ClassB. The lifetime of ClassA depends on the lifetime of ClassB.

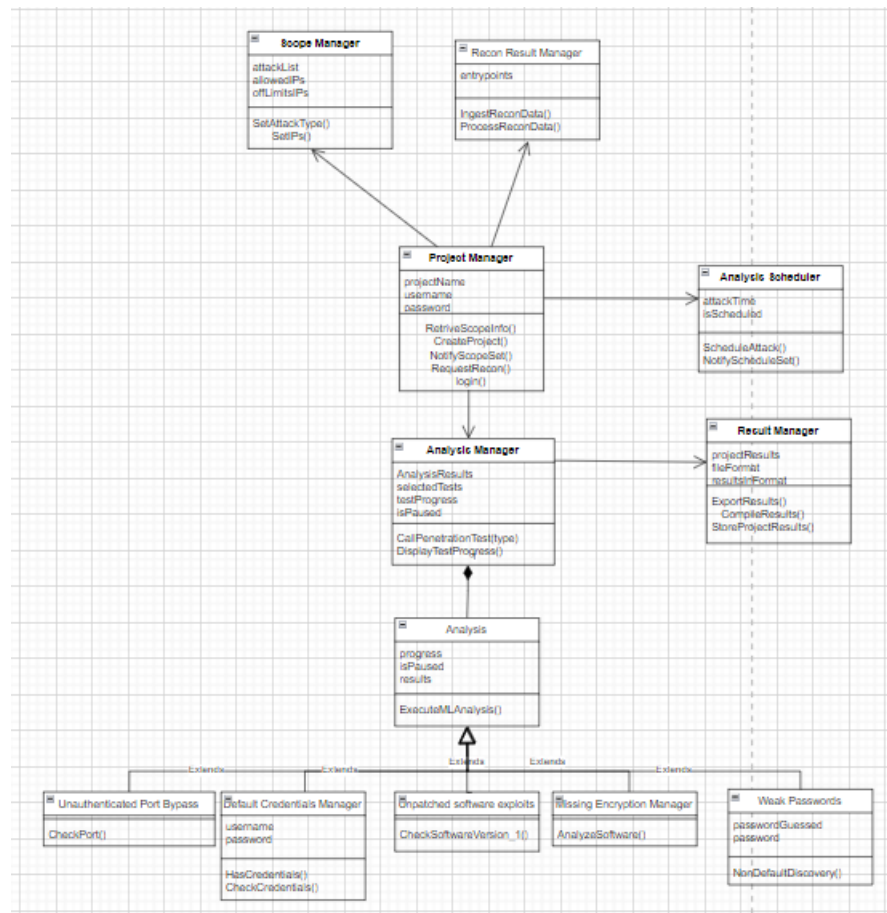


Figure 12. Class Diagram for the INFILTR8 system

3.2.2.1. Project Manager Class

[SRS 3.2.2.1.1] The Scope Manager shall store the attributes as defined in Table 3.2.2.1.-1.

Table 3.2.2.1.-1: Attributes of the Project Manager Component

Req Number	Attribute	Values and Constraints	Description
1	projectName	Required; Editable; Str	Name of the INFILTR8 project
2	username	Required; Editable; Str	A name used to identify a user and is part of the user account
3	password	Required; Editable; Str[15-255]	a string of characters used to verify the identity of a user and is part of the user account

[SRS 3.2.2.1.2] The project manager shall allow the Analyst to create a new user account.

[SRS 3.2.2.1.3] The user account shall include a username and a password.

[SRS 3.2.2.1.4] The project manager shall inform the Analyst about the CAT 1 STIG constraints to set a new password.

[SRS 3.2.2.1.5] The project manager shall ask the Analyst for a new password.

[SRS 3.2.2.1.6] The project manager shall verify that the new password includes:

- a) at least one upper-case character
- b) at least one lower-case character
- c) at least one numeric character
- d) at least one special character
- e) at least 15 characters in length.

Refer to STIG requirements V-71903, V-71905, V-71907, V-71909, V-71935

[SRS 3.2.2.1.7] The project manager shall allow the Analyst with a user account to login by providing a username and a password.

[SRS 3.2.2.1.8] The project manager shall verify provided Analyst's username and password against saved values.

[SRS 3.2.2.1.9] The project manager shall allow the Analyst up to three attempts to provide the correct username and password.

[SRS 3.2.2.1.10] The project manager shall lock the Analyst account after three failed attempts to login.

[SRS 3.2.2.1.11] The project manager shall allow the Administrator to unlock a locked Analyst account.

[SRS 3.2.2.1.12] The project manager shall allow the Analyst to use INFILTR8 after a successful login attempt (provided Analyst's username and password match saved values).

[SRS 3.2.2.1.13] The project manager shall allow the Analyst to open an existing project by selecting the name of the project.

[SRS 3.2.2.1.14] The project manager shall retrieve the opened project's data.

[SRS 3.2.2.1.15] The project manager shall allow the Analyst to create a new project.

[SRS 3.2.2.1.16] The project manager shall allow the Analyst to enter the name of the project.

[SRS 3.2.2.1.17] After providing the name of the project, the project manager shall direct the application flow to ingest reconnaissance result files and create Valid Entry Point List.

[SRS 3.2.2.1.18] After ingesting reconnaissance result files, the project manager shall direct the application flow to set the scope of the project.

[SRS 3.2.2.1.20] After generating the enhanced AI Valid Exploit List, the project manager shall direct the application flow to schedule the analysis if desired.

[SRS 3.2.2.1.21] If an analysis is not scheduled, the project manager shall direct the application flow to conduct the analysis.

3.2.2.2. Recon Result Manager Class

[SRS 3.2.2.2.1] The Recon Result Manager shall store the attributes as defined in Table 3.2.2.2.-1.

Table 3.2.2.2.-1: Attributes of the Recon Result Manager Component

Number	Attribute	Values and Constraints	Description
1	<i>ValidEntrypoints</i>	Required; Editable; Str List	Valid entrypoints that are available on the network found in the reconnaissance result files

[SRS 3.2.2.2.2] The Recon Result Manager shall prompt the analyst to select the folder where the reconnaissance result files from Nessus are.

[SRS 3.2.2.2.3] The Recon Result Manager shall not require all the reconnaissance result files from Nessus to allow the Analyst to proceed.

[SRS 3.2.2.2.4] The Recon Result Manager shall inform the analyst if any of the reconnaissance result files cannot be accessed or are not present.

[SRS 3.2.2.2.5] The Recon Result Manager shall ingest the reconnaissance result files from the selected folder and create a list of valid entrypoints that are available on the network.

[SRS 3.2.2.2.6] The Recon Result Manager shall inform the analyst of any errors while ingesting the reconnaissance result files.

3.2.2.3. Scope Manager Class

[SRS 3.2.2.3.1] The Scope Manager shall store the attributes as defined in Table 3.2.2.3.-1.

Table 3.2.2.3.-1: Attributes of the Scope Manager Component

Number	Attribute	Values and Constraints	Description
1	entryPointList	Required; Editable; List of type of entry points chosen.	A list of possible entry points chosen by the analyst to be performed during the analysis.
2	allowedIPs	Required; Editable; List of IP addresses	A list of IP addresses specified by the analyst on which the penetration testing should be done.
3	offLimitsIPs	Required; Editable; List of IP addresses	A list of IP addresses specified by the analyst on which the penetration testing should not be done.

[SRS 3.2.2.3.2] The Scope Manager shall allow the analyst to select the following potential entry point attack types to be performed:

- Unauthenticated port bypass,
- default credentials,
- unpatched software exploits,
- missing encryption protocols,
- weak passwords (brute force),

[SRS 3.2.2.3.3] The Scope Manager shall allow the analyst to write the list of the allowed IP addresses that undergo analysis.

[SRS 3.2.2.3.4] The Scope Manager shall include the allowed IP addresses for analysis.

[SRS 3.2.2.3.5] The Scope Manager shall allow the analyst to write the off-limits list of IP addresses.

[SRS 3.2.2.3.6] The Scope Manager shall exclude the off-limits list of IP addresses for analysis.

[SRS 3.2.2.3.7] The Scope Manager shall allow the analyst to set IP addresses using the Classless Inter-Domain Routing (CIDR) format, e.g., 192.168.1.0/16.

[SRS 3.2.2.3.8] The Scope Manager shall allow the analyst to upload a file with the list of the allowed IP addresses.

[SRS 3.2.2.3.9] The Scope Manager shall allow the analyst to upload a file with the off-limits list of IP addresses.

3.2.2.4. Log Manager Class

[SRS 3.2.2.5.1] The Log Manager shall store the attributes as defined in Table 3.2.2.5.-1.

Table 3.2.2.5.-1: Attributes of the Log Manager Component

Number	Attribute	Values and Constraints	Description
1	<i>analystActions</i>	Required; Editable; Str List	A list of commands requested by the analyst in a sequential order
2	<i>systemActions</i>	Required; Editable; Str List	A list of commands executed by the analyst in a sequential order

[SRS 3.2.2.5.2] The Log Manager shall append the analyst's actions to a log.

[SRS 3.2.2.5.3] The Log Manager shall append the system's actions to a log.

▪ Analysis Scheduler Class

[SRS 3.2.2.6.1] The Analysis Scheduler shall store the attributes as defined in Table 3.2.2.6.-1.

Table 3.2.2.6.-1: Attributes of the Analysis Scheduler Component

Number	Attribute	Values and Constraints	Description
1	<i>Analysis Time</i>	Required; Editable; Date-Time	The date and time at which the Analysis is scheduled to start
2	<i>isScheduled</i>	Required; Editable; Boolean	The value is true when a penetration testing is scheduled and false otherwise

[SRS 3.2.2.6.2] The Analysis Scheduler shall allow the analyst to schedule an Analysis at the specified date and time, see section 3.1.1 User Interfaces Figure 7.

[SRS 3.2.2.6.3] The Analysis Scheduler shall schedule an Analysis at the specified date and time.

[SRS 3.2.2.6.4] The Analysis Scheduler shall notify the analyst when an Analysis has been scheduled.

[SRS 3.2.2.6.5] The Analysis Scheduler shall start Analysis at the scheduled time.

3.2.2.5. Result Manager Class

[SRS 3.2.2.7.1] The Result Manager shall store the attributes as defined in Table 3.2.2.7.-1;

Table 3.2.2.7.-1: Result Manager Component

Req Number	Attribute	Values and Constraints	Description
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1	projectResults	Required; Editable; File	A file containing the compiled results of the analysis.
2	fileFormat	Required; Editable; String	The file format that was specified by the analyst to export the results.
3	resultsInFormat	Required; Editable; File	A file in the selected file format by the users, that contains the results from the analysis.

[SRS 3.2.2.7.2] The Result Manager shall export the results of the analysis in the file format that is specified by the analyst.

[SRS 3.2.2.7.3] The Result Manager shall compile the analysis results.

[SRS 3.2.2.7.4] The Result Manager shall store the project results at the directory location specified by the analyst.

3.2.3. Stimulus

[SRS 3.2.3.1] When creating a project, if the user has not uploaded reconnaissance files, the system shall prevent the user from creating and running the project.

3.2.3.1. User Interfaces

Settings screen

[SRS 3.2.3.1.1] When on the Settings screen, if the font size button is pressed, the system shall include additional font sizes for the user.

[SRS 3.2.3.1.2] When on the Settings screen, if the dark mode button is pressed, the system shall include additional font colors for the user.

[SRS 3.2.3.1.3] When on the Settings screen, if the help button is pressed, the system shall display the purpose and main functionalities of the system.

Login Screen

[SRS 3.2.3.1.4] When selecting the “Login” button, the system shall allow the analyst access to the system if valid username and password combinations are provided.

[SRS 3.2.3.1.5] When selecting the “Forgot Password?” button, the system shall display a pop-up screen to the analyst to change their password.

Testing Screen

[SRS 3.2.3.1.6] When selecting an entry point and a start time, the system shall schedule an analysis at the selected time.

[SRS 3.2.3.1.7] When selecting the “View Result” button, the system shall display the results.

Dashboard Screen

[SRS 3.2.3.1.8] When selecting the “Create Project” button, the system shall display the “Create New Project” window, see Section 3.1.1.

[SRS 3.2.3.1.9] When selecting the “Open Existing Project” button, the system shall display the file explorer window of the computer.

[SRS 3.2.3.1.10] When selecting the “Sync Project” button, the system shall display the “Sync Project” window, see Section 3.1.1.

[SRS 3.2.3.1.11] When selecting the “Discard all” button, the system shall discard all the files on the system.

[SRS 3.2.3.1.12] When selecting a Notification, the system shall display a pop-up window to display the text and remove the “Unread” label from the notification.

[SRS 3.2.3.1.13] When selecting the “Select Files” button, the system shall display the file explorer window of the computer.

[SRS 3.2.3.1.14] When selecting the “x” selection at the top right corner of a file, the file shall be removed from the system.

Project Screen

[SRS 3.2.3.1.15] When selecting a project folder, the system shall load the selected project.

[SRS 3.2.3.1.16] When selecting the “Start Analysis” button, the system shall begin the analysis on the defined scope.

Report Screen

[SRS 3.2.3.1.17] When selecting the drop-down “Format to export”, the system shall display the options for the file format of the results.

[SRS 3.2.3.1.18] When selecting the “Export” button, the system shall export the results of the analysis in the selected format.

Collapsable Menu

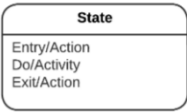
[SRS 3.2.3.1.19] When selecting one of the options seen on the drop-down menu, the system shall redirect the analyst to the selected page.

3.2.3.2. State Transitions

The system states define the different ways the system behaves by responding to specific events when certain conditions are met. The available functionality and the way the system provides the functionality may change based on the state of the system at that moment.

The legend of the state transition diagrams is defined in Table 3.2.3-1: State Transition Diagram Legend.

Table 3.2.3.1: State Transition Diagram Legend

NAME	NOTATION	DESCRIPTION
State		Defines the system's behavior at the given point in time. It may include entry or exit action and an activity.
Final State		Start of a state transition diagram.
Initial State		End of a state transition diagram.
Transition		State transition from State A to State B may occur when Event A occurs, and the Condition (optional Boolean expression) is satisfied. The action (optional activity) is performed only if the transition is done.

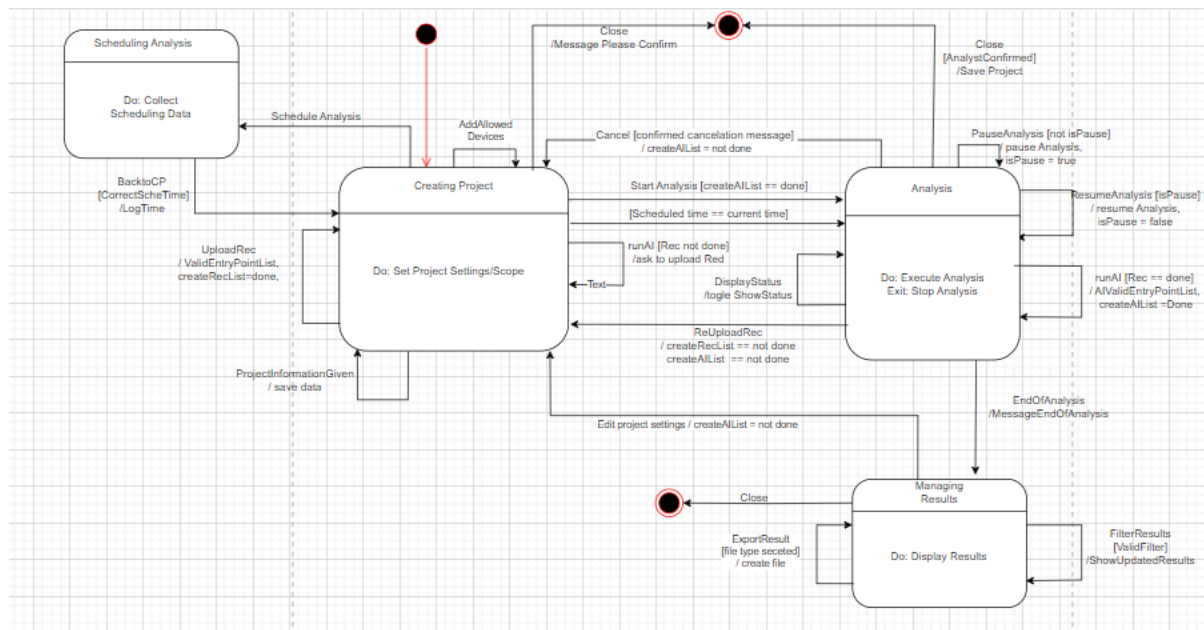


Figure 13. INFILTR8 State Transition Diagram.

[SRS 3.2.3.2.1] INFILTR8 shall include Creating Project, Scheduling Analysis, Analysis and Managing Results.

[SRS 3.2.3.2.2] Creating Project state shall be the start state.

[SRS 3.2.3.2.3] When in Creating Project, Analysis, or Managing Results states, if the close system event occurs, the system shall transition to the end state.

[SRS 3.2.3.2.4] When in the Creating Project state, if the reconnaissance files are uploaded, the system shall set variable rec to done and generate the *Valid Entrypoint List*.

[SRS 3.2.3.2.6] When in the Creating Project state, if the runAI button is pressed and the reconnaissance is not done, then the system shall send message to upload reconnaissance files first.

[SRS 3.2.3.2.7] When in the Creating Project state, if the analyst provides project information, the system shall save it.

[SRS 3.2.3.2.8] When in the Creating Project state, if AddAllowedDevices is pressed, the system shall allow the analyst to add a file or input an IP list of allowed devices.

[SRS 3.2.3.2.9] When in the Creating Project state, if the Start Analysis option is selected and createAllList is done, then the system shall transition to the Managing Results state.

[SRS 3.2.3.2.10] When in the Creating Project state, if the scheduled time is equal to the current time, then the system shall transition to the Analysis state.

[SRS 3.2.3.2.11] When in the Creating Project state, if the Schedule Analysis option is pressed the system shall transition to the Scheduling Analysis state.

[SRS 3.2.3.2.12] When in the Scheduling Analysis state, if the correct scheduling data is provided then the system shall save the scheduled time and transition to the Creating Project state.

[SRS 3.2.3.2.14] When in the Analysis state, if the Pause option is selected and isPause is false, then the system shall pause the analysis being done towards all allowed devices and set isPause to true.

[SRS 3.2.3.2.15] When in the Analysis state, if the Resume option is selected and isPause is true, then the system shall resume the analysis in all allowed devices and set isPause to false.

[SRS 3.2.3.2.16] When in the Analysis state, if the DisplayStatus option is pressed and isDisplayed is true (false), then the system shall *set isDisplay to false and stop displaying (set isDisplaying to true and start displaying)* the analysis status to the analyst.

[SRS 3.2.3.2.17] When in the Analysis state, if the Cancel option is selected and cancelation message is confirmed, then the system shall transition to Creating Project state and set createAIList to not done.

[SRS 3.2.3.2.18] When in the Analysis state, if the end of the analysis is reached, the system shall transition to the Managing Results state and display message “Analysis Complete”.

[SRS 3.2.3.2.19] When in the Managing Results state, if the button Filter Results is pressed, then the system shall filter the results and display them to the analyst.

[SRS 3.2.3.2.20] When in the Managing Results state, if the Export results option is pressed and a file type is selected, then the system shall create the results in the selected file type.

[SRS 3.2.3.2.21] When in the Managing Results state, if the Edit project settings is pressed, then the system shall transition to the Creating Project state and set createAIList to not done.

3.3. Non-behavioral Requirements / Quality Attributes

This section describes performance, availability, security, maintainability, and portability requirements. The performance requirements give a qualitative description of the system's process execution time.

3.3.1. Performance Requirements

[SRS 3.3.1.1] INFILTR8 shall display a warning message when the use of the CPU is above 80.

[SRS 3.3.1.2] INFILTR8 shall display a warning message when the use of memory is above 80.

[SRS 3.3.1.3] INFILTR8 shall display a warning message when the use of disk space is above 80.

3.3.2. Availability

[SRS 3.3.2.1] INFILTR8 shall use a recovery procedure to resume the analysis and the data in the event of a failure.

[SRS 3.3.2.2] INFILTR8 shall use checkpoints to save data and the state of the system to support the recovery procedure.

3.3.3. Security

This section specifies the factors that shall protect the software from accidental or malicious access, use, modification, destruction, or disclosure.

[SRS 3.3.3.1] INFILTR8 shall verify validity periods on all application messages using WS-Security or SAML assertions to prevent the use of outdated security tokens.

[SRS 3.3.3.2] INFILTR8 shall require analysts to provide login information, including their initials and a password with at least 15 characters of length, as stated in CAT 1 Vulnerabilities, V-222522, and V-222536.

[SRS 3.3.3.3] INFILTR8 shall maintain user-activity logs, including login attempts, access to sensitive functions, and modification of system configurations.

[SRS 3.3.3.5] INFILTR8 shall allow analysts to view user-activity logs.

[SRS 3.3.3.6] INFILTR8 shall use role-based access control (RBAC) mechanisms to ensure that users can only access functionalities and data that are relevant to their roles and responsibilities.

[SRS 3.3.3.7] INFILTR8 shall employ algorithms or checksums to verify the integrity of data at various checkpoints throughout the system's operation.

[SRS 3.3.3.8] INFILTR8 shall meet the requirements of the 32 CAT I STIG vulnerabilities that are applicable and listed in https://www.stigviewer.com/stig/application_security_and_development/ (see Table 3.3.3-1 for applicable vulnerabilities).

3.3.4. Maintainability

[SRS 3.3.4.1] INFILTR8 shall use version control systems to manage and document any code changes and keep track of progressive developments.

[SRS 3.3.4.2] INFILTR8 shall be designed with modular components that allow for easy replacement or updating of individual modules.

[SRS 3.3.4.3] All third-party dependencies shall be documented, including version, source, and any modifications made. This documentation shall also include guidelines for updating these dependencies and assessing the potential impacts of such updates on the system.

[SRS 3.3.4.3] INFILTR8 code's repository shall contain a markdown file that justifies the 32 CAT 1 STIG vulnerabilities.

[SRS 3.3.4.4] INFILTR8 markdown file shall include a table that addresses the vulnerability ID, description, and mitigating reason for each vulnerability as shown in Table 3.4.3-1.

3.3.5. Portability

[SRS 3.3.5.1] INFILTR8 shall ensure that no more than 10% of its components contain host-dependent code to facilitate easier portability and reduce platform-specific dependencies.

[SRS 3.3.5.2] INFILTR8's total codebase shall contain no more than 5% of host-dependent code.

This minimizes reliance on any single host's operating system or hardware specifications, enhancing system portability across different environments.

[SRS 3.3.5.3] INFILTR8 shall standardize the use of cross-platform third-party libraries, such as OpenSSL for cryptography.

3.3.6. Design and Implementation Constraints

[SRS 3.3.6.1] INFILTR8 shall be developed using SvelteKit 2.5.24 for the front-end and Python 3.11.4 or Node.js v20.17.0 for the back-end.

[SRS 3.3.6.2] INFILTR8's back-end shall utilize Neo4J DBMS.

[SRS 3.3.6.3] INFILTR8's memory usage shall not exceed 6GB of RAM.

[SRS 3.3.6.4] INFILTR8 shall be capable of running on machines with a minimum of 4-cores and 2.5GHz CPU speed.

[SRS 3.3.6.5] INFILTR8 storage usage shall not exceed 20GB of available storage.

3.4. Other Requirements

3.4.1. Data Flow

The system receives information (data inputs) from external entities and transforms them into data outputs consumed by external entities. Figure 14 shows the Context Diagram (also known as Data Flow Diagram (DFD) L0) of the INFILTR8 with its data inputs and outputs. A DFD L1, see Figure 3.4.1-1, shows internal process transformations that convert data inputs from either external entity, other process transformation(s), or data storage(s) into data outputs that feed other process transformation(s), data storage(s) or external entities. Information about data terms can be seen in the data dictionary on Section 3.4.2.

3.4.1.1. Context Diagram (AKA DFD Level 0)

Table 3.4.1-1 displays the legend for the context diagram.

Table 3.4.1-1: Context Diagram Legend

NAME	NOTATION	DESCRIPTION
System		The system being modeled. Represented as a single process that takes input data and transforms it into output data.
Terminator		External entity that interacts with the system by sending or receiving data. Also known as source or sink.
Data Flow		Represents data or information that flows between the system and a terminator or vice versa. The name of a data flow may be followed by an apostrophe – i.e. <data_name>'. This signifies that the data is altered in some way.

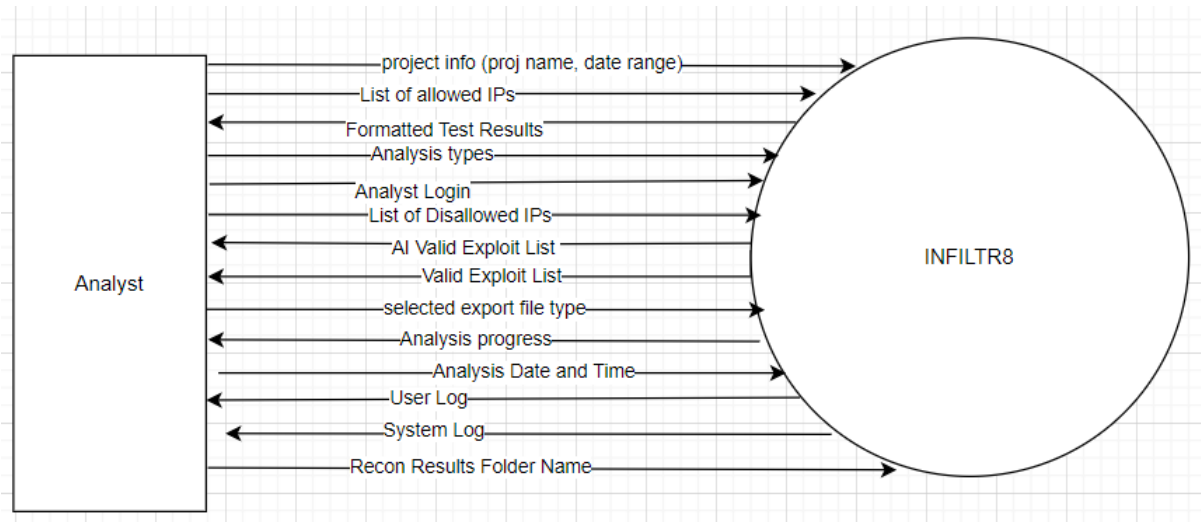


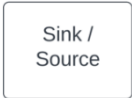
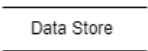
Figure 3.4.1-1. INFILTR8 Context Diagram

3.4.1.2. Data flow diagrams

Table 3.4.1.2 displays the legend for the data flow diagrams.

Table 3.4.1-2: Data Flow Diagram Legend

NAME	NOTATION	DESCRIPTION
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Process		Takes input data and transforms it into resulting output data. Also known as Data Transformation. It has a process number useful to keep track of the diagram level.
Sink		Consumes information produced by the system. Also known as a Terminator or External Entity.
Source		Produces information used by the system. Also known as a Terminator or External Entity.
Data Store		Stores data persistently for later use.
Data Flow		Represents data or information that flows between process-process, process-data store, process-sink, process-source. The name of a data flow may be followed by an apostrophe – i.e. <data_name>' when the data altered in some way.

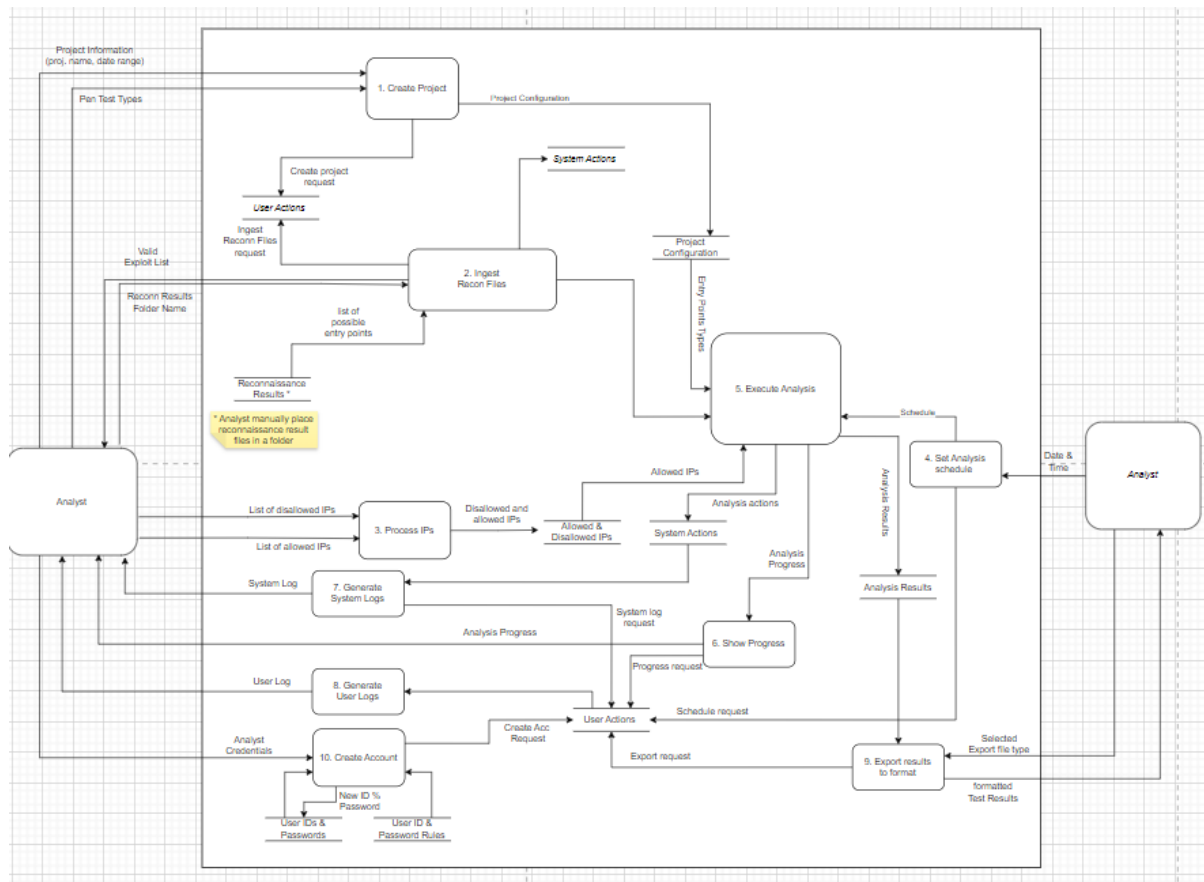


Figure 3.4.1-2. INFILTR8 Data Flow Diagram

[SRS 3.4.1.2.1] The “Create Project” process shall receive data inputs *Entry Point Types* and *Project Information*, and produce outputs *Project Configuration* and *Create Project Request* by the following steps:

1. Gather project information
2. Gather selection of Entry Point types
3. Create new and store project configuration
4. Log new project request to the User Actions.

[SRS 3.4.1.2.2] The “Ingest Reconnaissance Files” process shall receive data inputs *Reconnaissance Results Folder Name* and *List of Possible Entry Points*, and produce output *Valid Entrypoint List* and *Ingest Reconnaissance Files Request* by the following steps:

1. Gather the Reconnaissance Results Folder name.
2. Collect List of Possible Entry Points from Reconnaissance Files.
3. Apply the Machine Learning algorithm that uses linear regression to analyze the data.
4. Apply the metrics to the data to give it a weight.
5. Create and store the Valid Entrypoint List.
6. Log Process Reconnaissance Files request to the User Actions

[SRS 3.4.1.2.3] The “Process IPs” process shall receive data inputs *List of Disallowed IPs* and *List of Allowed IPs* and produce output *Disallowed and Allowed IPs* by the following steps:

1. Gather the disallowed IPs.
2. Gather the allowed IPs.
3. Create and store a list of the disallowed and allowed IPs.

[SRS 3.4.1.2.4] The “Set Analysis Schedule” process shall receive the data input *Date and Time* and generate outputs *Schedule* and *Schedule request* by the following steps:

1. Gather the Date and Time for scheduling the analysis.
2. Generate a schedule with the given Date & Time.
3. Log Schedule request to the User Actions.

[SRS 3.4.1.2.5] The “Execute Analysis” process shall receive the **data inputs** *Entry Point Types*, *Allowed IPs*, *Schedule*, *Analysis Actions*, *Analysis Progress*, *Analysis Results*, by the following steps:

1. Gather the entry point types.
2. Gather the Allowed IPs.
3. Gather the schedule.
4. Begin Analysis of endpoints leveraging Artificial Intelligence powered algorithms based on Entrypoint Types that were selected.
5. Gather the crash notifications
6. Create a log for every action done by the system
7. Continuously report Analysis Progress
8. Store test results of the analysis

[SRS 3.4.1.2.6] The “Show Progress” process shall receive the data input *Analysis Progress* and generate outputs *Analysis Progress* and *Progress request* by the following steps:

1. Gather the Analysis information.
2. Display the progress to the analyst.
3. Log Show Progress request to the User Actions.

[SRS 3.4.1.2.7] The “Generate System Log” process shall receive the data input *System Actions* and generate the data outputs *List of System Actions* and *Generate System Log request*.

[SRS 3.4.1.2.8] The “Generate User Logs” process shall receive data inputs *User Actions* and produce output *List of User Actions* and *Generate User Logs request*.

[SRS 3.4.1.2.11] The “Export Results to Format” process shall receive the data input *Test Results* and *Selected Export file Type* and produce the data outputs *Export Request* and *Formatted Test Results* by the following steps:

1. Gather the selected Export file type.
2. Format the test results into the selected file type.
3. Log Export request to the User Actions

[SRS 3.4.1.2.9] The “Create Account” process shall receive the data input *Analyst Credentials, User ID & Password Rules, User IDs & Passwords*, and generate outputs *New User ID & Password* and *Create Account request* by the following steps:

1. Gather Analyst Credentials.
2. Verify Analyst Credentials meet User ID & Password Rules.
3. Verify User ID of the Analyst Credentials is not in the User IDs & Passwords storage
4. Log Create Account request to the User Actions.

3.4.1.3. Data Dictionary

+	Sequence - Represents concatenation of multiple pieces of data
[]	Selection - The data within here is optional
{ }	Repetition - The data within here may be repeated
	Or - The data on either side of this symbol is acceptable

Figure 3.4.1.2-1. Data Dictionary Legend

Table 3.4.1.2-1: Data Dictionary

Composite / Atomic Term Name	Composite formula / Atomic Value	Description	Type	Default Value
List of Allowed IPs	{IP}	A list containing the IP addresses of the devices that will undergo analysis.	List	Null
List of Disallowed IPs	{IP}	A list containing the IP addresses of the devices that will not undergo analysis.	List	Null
Allowed IPs	{IP}	A list containing the IP addresses that can undergo analysis.	List	Null
Project Information	= ProjectName + DateRange	The information necessary to create a project.	String, date, date	Null
ProjectName	{Ch}^30	The name	String	Null

		provided by the analyst to identify the project.		
DateRange	date	The range of dates the project is expected to remain active.	Date	Null
Formatted Test Results	List (XML or PDF)	Test results of the penetration testing. Formatted in a list and are exported in either an XML or PDF file.	XML or PDF file	Null
analysis Types	{analysisType}	List of selected analysis types that the analyst wants to implement on the allowed devices.	List of analysis types	Null
Selected Export File Type	{Ch}^5	Selection of PDF or XML format for the exports of the final analysis report.	File extension to determine if	Null
analysis Progress	None	A display of the progress of the analysis to the analyst.	Visual representation of the progress of the analysis.	Null
Test Progress	None	The current progress of the analysis is used to generate an update for the analyst.	Tuple	Null
Analysis Date and Time	= Date + Time	The specification of when a project will take place when it is being scheduled.	The date and time set by the analyst	Null
User Log	= {Char}^120	A log is generated by the	Text file	Null

		system that tracks the actions taken by the user.		
System Log	= {Char}^120	A log is generated by the system that tracks the actions taken by the system.	Text File	Null
Recon Results Folder Name	{Char}^256	The name of the folder where the results of the project are saved.	String	Null
Shell access, error/crash notification	{Char}^30	Notification to the analyst of an error or a system crash	String	Null
Create Project Request	= {Char}^30	A request generated by the system to add to the user logs when there is a request to view the progress of an ongoing analysis.	String	Null
Project Configuration	= {Char}^30 + date	The name and date range for the project as set by the analyst in charge of the project.	String + date	Null
Entry Point Types	= {Char}^30	A list containing the selected penetration testing techniques selected by the analyst to perform on the allowed devices. The available EntryPointTypes are unauthenticated port bypass, default	List of strings	Null

		credentials (specific services), unpatched software exploits, plain text missing encryption in protocols (detecting and using password exchanges), weak passwords (brute force)		
List of Possible Entry points	= {IP}	The ranked list of valid entry points available on the system as specified by the reconnaissance files.	List	Null
Valid Entrypoint List	= {IP}	The list of possible entry points as specified by the AI.	List	Null
Disallowed and allowed IPs	= {IP}	A list containing all devices accessible by the system. It contains both the disallowed devices and the allowed devices. It will be processed by the INFILTR8 system to perform the analysis only on the allowed devices.	List	Null
System Actions	= {Char}^120	System actions as they are performed during the analysis. This	Text file	Null

		will be sent to a log of all of the system actions.		
Schedule	= time + date	The time and date specified by the analyst during which they want to perform an analysis.	Time and date	Null
Schedule Request	= {Char}^30	A request that is generated by the system when an analyst submits the schedule for a analysis.	String	Null
Export Request	= {Char}^30	A request that is generated by the system for the user actions log when an analyst needs to export the analysis results.	String	Null
Create acc Request	= {Char}^30	A request generated by the system when an analyst tries to create an account for the first time.	String	Null
Analyst Credentials	= {Char}^30	A username and password provided by the analyst when creating an account for the first time.	List	Null
Testing Actions	= {Char}^30	The action taken by the AI to be used to generate the system action log.	List	Null
Analysis Results	= {String}	The results of the penetration test	List	Null

		that was run by the analyst.		
New ID & Password	= {String}	The list containing the username and password generated by the analyst.	List	Null
User IDs & Password Rules	= {Char}^30	The database that stores the rules for new passwords and usernames.	List	Null
User Actions	= {Char}^30	The database that stores the log of the actions done by the analyst throughout the analysis.	List	Null
System log request	{Char}^30	A request generated to log the actions of the analyst.	List	Null
Progress Request	{Char}^30	A request generated to log the actions of the analyst when they see the progress of the analysis.	List	Null
Process Recc Files Request	{Char}^30	A log is generated by the system to record the actions of the analysts when they process the reconnaissance files.	List	Null

3.4.2. Operations

[SRS 3.1.17.1] The INFILTR8 system shall support various modes of operation to facilitate the Analysis efficiently, including:

[SRS 3.1.17.2] User-friendly Interface: INFILTR8 shall have an intuitive interface to make it a tool that is easy to use for analysts.

[SRS 3.1.17.3] Real-Time Progress Tracking: Analysts shall have the ability to monitor the progress of an ongoing analysis in real-time.

[SRS 3.1.17.4] Customizable Scopes: The INFILTR8 system shall allow the analyst to configure the scope more than once to add more allowed or disallowed devices if necessary.

3.4.3. CAT 1 STIG Vulnerabilities

Table 3.4.3-1: Justification of CAT 1 STIG Vulnerabilities

VULNERABILITY ID	DESCRIPTION	MITIGATING REASON
V – 222400	Validity periods must be verified on all application messages using WS-Security or SAML assertions.	This is a security measure to prevent the use of outdated or expired security tokens.
V – 222404	The application must use both the NotBefore and NotOnOrAfter elements or OneTimeUse element when using the Conditions element in a SAML assertion.	For this project this would not apply.
V – 222612	The application must not be vulnerable to overflow attacks.	Overflow attacks, such as buffer overflows, can lead to serious security vulnerabilities, allowing attackers to overwrite memory and execute arbitrary code. INFILTR8 mitigates overflow attacks by implementing: 4. Input Validation 5. Buffer Size Checks 6. Secure Coding Practices 7. Memory Safety Languages 8. Static Code Analysis 9. Dynamic Analysis
V – 222578	The application must destroy the session ID value and/or cookie on logoff or browser close.	INFILTR8 erases the session ID value or cookies on logoff to ensure privacy, security and system integrity.
V – 222430	The application must execute without excessive account permissions.	INFILTR8 is used locally in a browser like Mozilla Firefox and doesn't connect to the internet. INFILTR8 manages session IDs and cookies to protect sensitive information and user sessions. INFILTR8 implements the following: 10. Session Management 11. Cookie Attributes ('HTTP Only', 'Same Site') 12. Session Timeout 13. Clear Cookies on browser close 14. Secure local storage

V – 222432	The application must enforce the limit of three consecutive invalid logon attempts by a user during a 15-minute time period.	INFILTR8 blocks the analyst from attempting to login after three consecutive invalid login attempts to ensure there is no unauthorized access to the system.
V – 222577	The application must not expose session IDs.	INFILTR8 implements measures to prevent session ID exposure by implementing: 15. Protect against session fixation 16. Session ID rotation 17. Session timeout 18. Use secure cookies
V – 222609	The application must not be subject to input handling vulnerabilities.	INFILTR8 address input handling vulnerabilities by implementing: 19. Avoid trusting user input 20. Use parameterized queries 21. File upload security (valid file types and restrict size) 22. Error Handling 23. Input Validation
V – 222608	The application must not be vulnerable to XML-oriented attacks	INFILTR8 mitigates XML-oriented attacks by implementing: 24. Avoid XML External Entity Attacks 25. Use a secure XML Parser 26. Input validation for XML Data
V – 222602	The application must protect from Cross-Site Scripting (XSS) vulnerabilities	INFILTR8 mitigates Cross-Site Scripting (XSS) vulnerabilities by implementing: 27. Output encoding 28. Content Security Policy (CSP) 29. Use frameworks with built-in protection (e.g. React, Angular, Vue...) 30. Escape Dynamic JavaScript values
V – 222601	The application must not store sensitive information in hidden fields	INFILTR8 does not store sensitive information in hidden fields by implementing these practices: 31. Avoid Storing Sensitive Data 32. Secure Communication Channels 33. Use local storage 34. Data Masking
V – 222607	The application must not be vulnerable to SQL Injection.	MongoDB is a NoSQL database and not susceptible to traditional SQL injection attacks. INFILTR8 prevents NoSQL injection by implementing: 35. Use parameterized queries 36. Sanitize input 37. Error Handling 38. Update dependencies

V – 222604	The application must be protected from command injection.	INFILTR8 mitigates command injection vulnerabilities, especially in the context of using MongoDB, by implementing: 39. Use parameterized queries 40. Sanitize input 41. Error Handling 42. Update dependencies
V – 222403	The application must use the NotOnOrAfter condition when using the SubjectConfirmation element in a SAML assertion.	The analysts shall not be required to log in to multiple applications, so this does not apply.
V – 222585	The application must fail to a secure state if system initialization fails, shutdown fails, or aborts fail.	INFILTR8 fails to a secure state by implementing: 43. Error handling 44. Graceful initialization 45. Shutdown process 46. Abort handling 47. Testing Scenarios
V – 222550	The application, when utilizing PKI-based authentication, must validate certificates by constructing a certification path to an accepted trust anchor.	INFILTR8 shall validate certificates as necessary to ensure system security and trustworthiness.
V – 222522	The application must uniquely identify and authenticate organizational users (or processes acting on behalf of organizational users).	INFILTR8 uses unique user IDs and passwords
V – 222554	The application must not display passwords/PINs as clear text.	INFILTR8 uses and stores passwords.
V – 222596	The application must protect the confidentiality and integrity of transmitted information.	INFILTR8 uses secure channels.
V – 222399	Messages protected with WS_Security must use time stamps with creation and expiration times.	N/A; INFILTR8 will not use WS_Security tokens.
V – 222658	All products must be supported by the vendor or the development team.	All products from INFILTR8 will be supported by the development team and vendor

V – 222659	The application must be decommissioned when maintenance or support is no longer available.	When decommissioned INFILTR8 will be available to terminate any files and storage
V – 222551	The application, when using PKI-based authentication, must enforce authorized access to the corresponding private key.	All PKI-Based Authentication shall give access to a corresponding private key.
V – 222620	Application web servers must be on a separate network segment from the application and database servers if it is a tiered application operating in the DoD DMZ.	INFILTR8 will have no use of the internet.
V – 222536	The application must enforce a minimum 15-character password length.	INFILTR8 makes use of a minimum 15-character password length.
V – 222643	The application must have the capability to mark sensitive/classified output when required.	INFILTR8 gives the option to mark sensitive data to specific users.
V – 222542	The application must only store cryptographic representations of passwords.	INFILTR8 store cryptographic representations of passwords.
V – 222543	The application must transmit only cryptographically-protected passwords.	INFILTR8 transmits cryptographically-protected passwords.
V – 222425	The application must enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	INFILTR8 gives the approved user specific data they are entrusted with.
V – 222642	The application must not contain embedded authentication data.	INFILTR8 requires analysts to give authentication details each time a user logs in.
V - 222662	Default passwords must be changed.	INFILTR8 requires analysts to change default passwords.
V - 222555	The application must use mechanisms meeting the requirements of applicable federal laws,	INFILTR8 will follow all the guidelines of Federal Laws, Executive Orders, Policies, Regulations, Stands for the cryptographic module.

	Executive Orders, directives, policies, regulations, standards, and guidance for authentication to a cryptographic module.	
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